Experiment No. 10

Aim: Explore the Cryptocurrency Landscape.

Theory:

1. Introduction to Cryptocurrency

Cryptocurrencies are digital or virtual currencies that use blockchain technology for secure and decentralized transactions. Unlike traditional currencies, they are not controlled by any central authority like banks or governments.

A blockchain is a distributed ledger where transactions are recorded in blocks and linked together using cryptographic techniques, ensuring:

- Transparency
- Security
- Immutability

2. Public vs Private Blockchain

Feature	Public Blockchain	Private Blockchain
Access	Open to everyone	Restricted access
Examples	Bitcoin, Ethereum	Hyperledger
Control	Decentralized	Controlled by organization
Transparency	High	Limited

3. Data Sources for Cryptocurrency Analysis

To understand the crypto market, data is collected from multiple sources:

- Financial Platforms
  Provide price, volume, market cap
  Example: CoinGecko, CoinMarketCap
- News Platforms
  Provide latest updates affecting crypto prices
  Example: CryptoPanic

- Social Media APIs
  Measure sentiment and trends
  Example: Twitter, Reddit, LunarCrush
- Regulatory Websites
  Provide legal frameworks and policies

4. Types of Cryptocurrencies

- Store of Value (SoV): Bitcoin
- Utility Tokens: Ethereum, Chainlink
- Stablecoins: USDT, USDC
- Governance Tokens: UNI, AAVE

5. Blockchain Technology Advancements

- Consensus Mechanisms:
  Proof of Work (PoW), Proof of Stake (PoS)
- Scalability Solutions:
  Layer-2 (Polygon), Sharding
- Interoperability:
  Cross-chain communication
- Privacy Features:
  Zero-Knowledge Proofs (ZKP)

6. Market Trends & Adoption

- Increasing use in digital payments
- Growth of DeFi (Decentralized Finance)
- Rising wallet users
- Increase in active blockchain addresses

Implementation:

code:

```
import requests
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

API_KEY = "fda6fb498508a25f987b1d9b25ddc927912e8e76ef516440a15fc964ebee75fe"
SYMBOL = "BTC"          # Bitcoin (you can use ETH, LINK etc.)
CURRENCY = "USD"
LIMIT = 100             # Last 100 days
```

STEP 1: FETCH DATA FROM API

```
def fetch_data():
    url = "https://min-api.cryptocompare.com/data/v2/histoday"

    params = {
        "fsym": SYMBOL,
        "tsym": CURRENCY,
        "limit": LIMIT,
        "api_key": API_KEY
    }

    response = requests.get(url, params=params)
    data = response.json()

    if data["Response"] == "Success":
        return data["Data"]["Data"]
    else:
        print("Error:", data.get("Message"))
        return None
```

STEP 2: PROCESS DATA

```
def process_data(raw_data):
    df = pd.DataFrame(raw_data)

    # Convert timestamp
    df['date'] = pd.to_datetime(df['time'], unit='s')
    df.set_index('date', inplace=True)

    # Rename columns
    df['price'] = df['close']
    df['volume'] = df['volumeto']
```

```
    # Calculations
    df['returns'] = df['price'].pct_change()
    df['MA7'] = df['price'].rolling(7).mean()
    df['MA30'] = df['price'].rolling(30).mean()
    df['volatility'] = df['returns'].rolling(7).std() * 100
    df['cumulative'] = (1 + df['returns']).cumprod()
    return df
```

STEP 3: VISUALIZATION

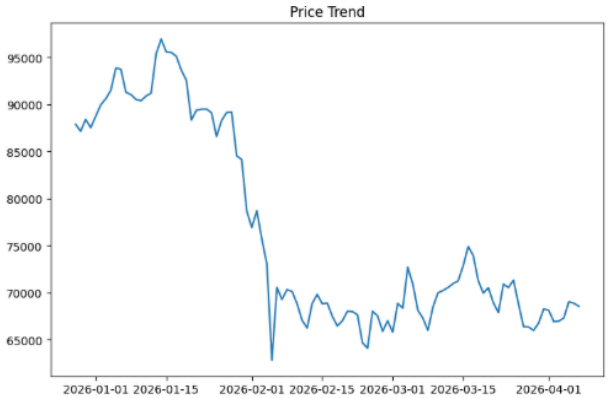
```
def plot_data(df):
    plt.figure(figsize=(15, 10))

    # Price Trend
    plt.subplot(2, 2, 1)
    plt.plot(df['price'])
    plt.title("Price Trend")

    # Moving Averages
    plt.subplot(2, 2, 2)
    plt.plot(df['price'], label="Price")
    plt.plot(df['MA7'], label="MA7")
    plt.plot(df['MA30'], label="MA30")
    plt.legend()
    plt.title("Moving Averages")

    # Volume
    plt.subplot(2, 2, 3)
    plt.bar(df.index, df['volume'])
    plt.title("Trading Volume")

    # Volatility
    plt.subplot(2, 2, 4)
    plt.plot(df['volatility'])
    plt.title("Volatility")
    plt.tight_layout()
    plt.show()
```

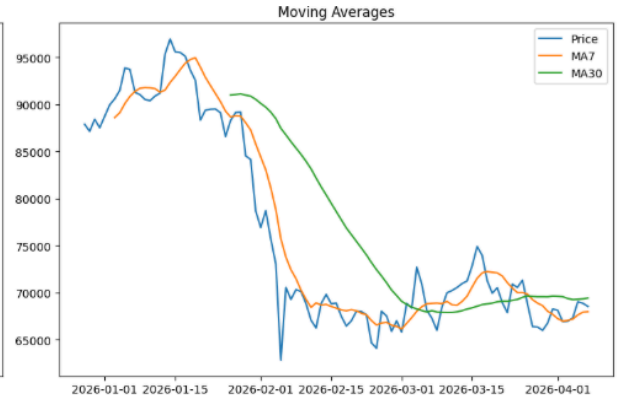
STEP 4: RUN PROGRAM

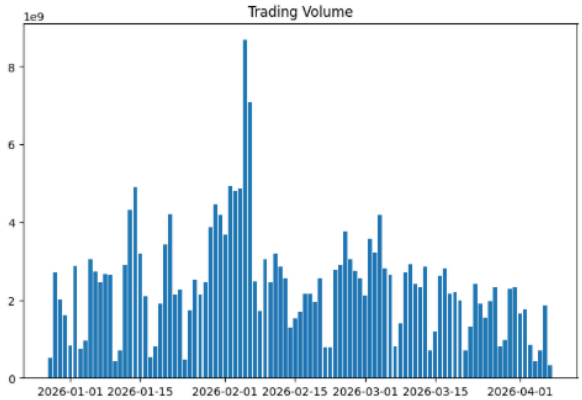
```
raw_data = fetch_data()

if raw_data:
    df = process_data(raw_data)
    print("\n=== DATA SAMPLE ===")
    print(df.head())
    plot_data(df)
```

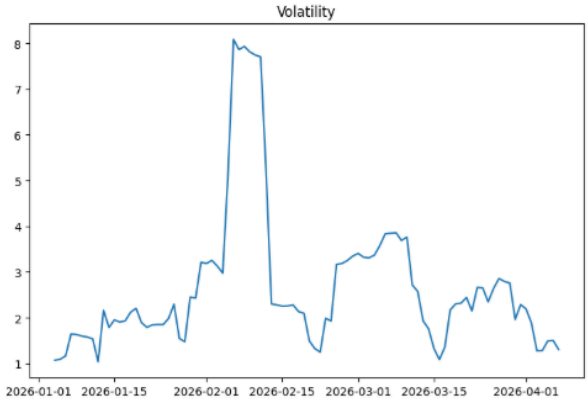
Output:

```
=== DATA SAMPLE ===
                  time      high       low      open  volumefrom  \
date
2025-12-28  1766880000  88018.43  87381.32  87823.31     5733.05
2025-12-29  1766966400  90327.16  86704.68  87886.81    30609.16
2025-12-30  1767052800  89341.34  86732.68  87133.26    22736.07
2025-12-31  1767139200  89105.68  87109.45  88410.54    18255.84
2026-01-01  1767225600  88806.59  87400.21  87519.13     9510.96

               volumeto     close conversionType conversionSymbol     price  \
date
2025-12-28  5.028765e+08  87886.81         direct                  87886.81
2025-12-29  2.698232e+09  87133.26         direct                  87133.26
2025-12-30  2.002631e+09  88410.54         direct                  88410.54
2025-12-31  1.607399e+09  87519.13         direct                  87519.13
2026-01-01  8.358194e+08  88742.68         direct                  88742.68

                 volume   returns  MA7  MA30  volatility  cumulative
date
2025-12-28  5.028765e+08       NaN  NaN   NaN         NaN         NaN
2025-12-29  2.698232e+09 -0.008574  NaN   NaN         NaN    0.991426
2025-12-30  2.002631e+09  0.014659  NaN   NaN         NaN    1.005959
2025-12-31  1.607399e+09 -0.010083  NaN   NaN         NaN    0.995816
2026-01-01  8.358194e+08  0.013980  NaN   NaN         NaN    1.009738
```

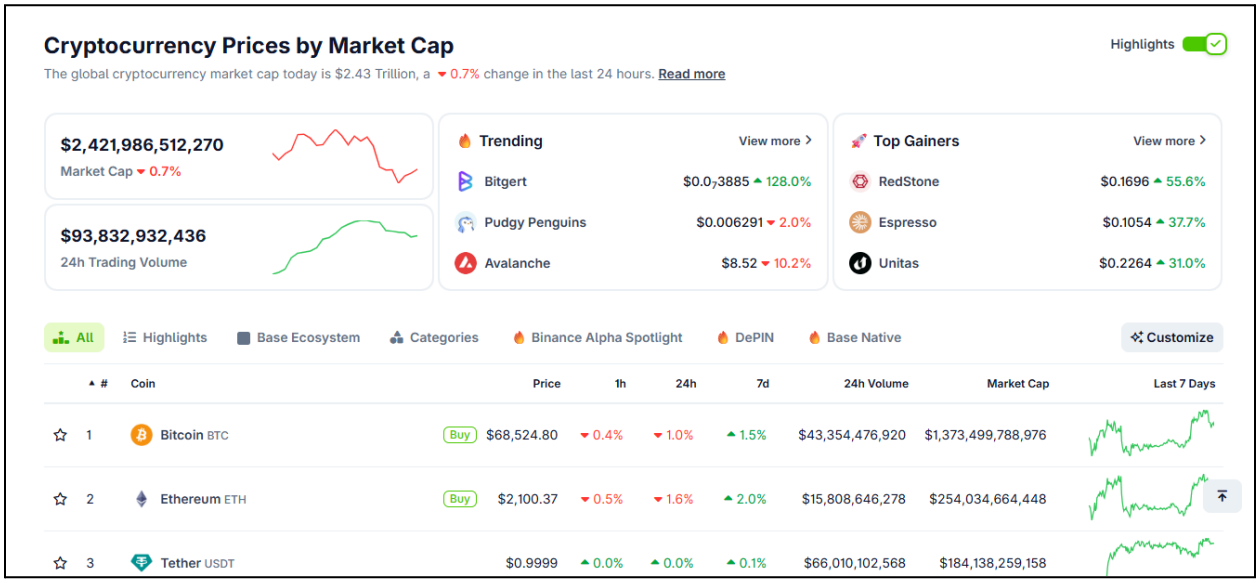








Coingecko website:



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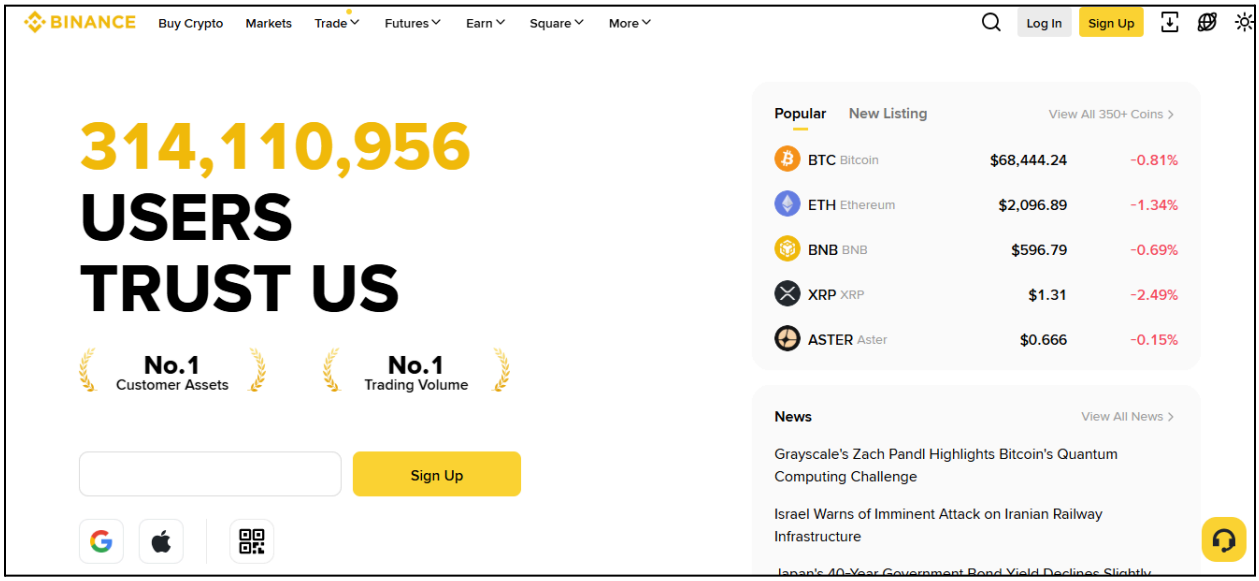

Conclusion:

A comprehensive understanding of the cryptocurrency ecosystem can be achieved by combining market data analysis, blockchain evaluation, and adoption metrics. This helps connect technical foundations with real-world trends and future developments.